

JARVIS – Just A Rather Very Intelligent Sorting Machine

ProfessionalGitHubDocumentation&ProjectReport

Abstract

JARVIS is a smart waste segregation system designed to automatically identify and sort waste materials using sensors, embedded electronics and mechanical actuation. The project aims to reduce contamination in recycling streams, improve sorting efficiency and support sustainable waste management.

Problem Statement

Manual waste segregation is time consuming, inconsistent and prone to contamination. Industries and municipalities require affordable intelligent systems capable of improving recycling rates.

Objectives

Automate waste segregation, reduce human effort, improve recycling efficiency, provide real-time feedback, support scalability and enable future AI integration.

Existing Features

Eco-friendly sorting, customizable operation, scalable design, reduced labor dependency, modular architecture and real-time monitoring.

Proposed Hardware

Arduino Uno, LCD display, servo motor, DC motor, sensors, breadboard, power supply, mechanical conveyor/chute structure and collection bins.

Software & Control Logic

Embedded C/C++ using Arduino IDE. Sensor readings are processed and corresponding actuators direct waste into appropriate bins.

Future Enhancements (Missing Information Added)

AI-based image classification, IoT dashboard, cloud analytics, mobile application, load-cell based weight tracking, machine vision and predictive maintenance.

Testing & Validation

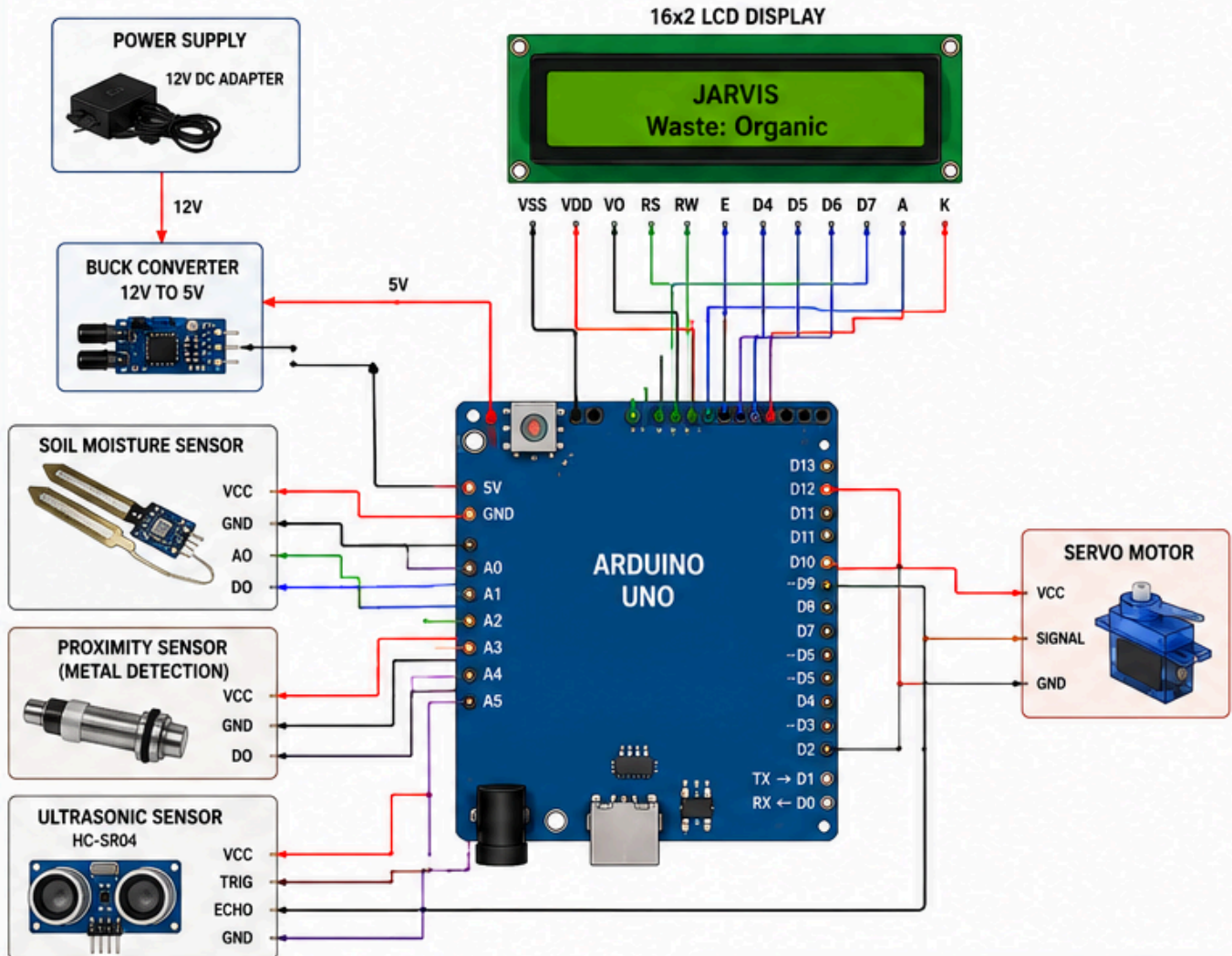
Sensor calibration, sorting accuracy tests, actuator response testing, power consumption measurements and long-duration operational trials.

Impact

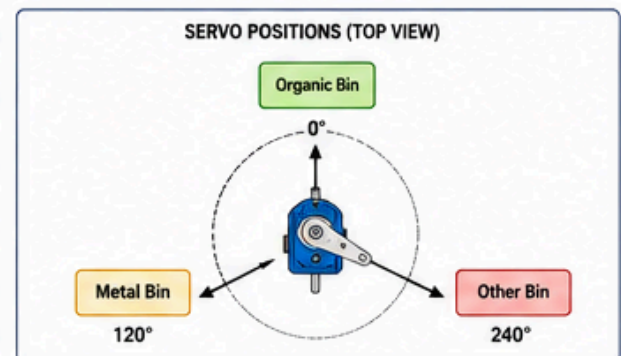
Improves recycling efficiency, reduces landfill waste, lowers operational cost and supports environmental sustainability.

CIRCUIT DIAGRAM – JARVIS SMART WASTE SEGREGATION SYSTEM

(Arduino Based)



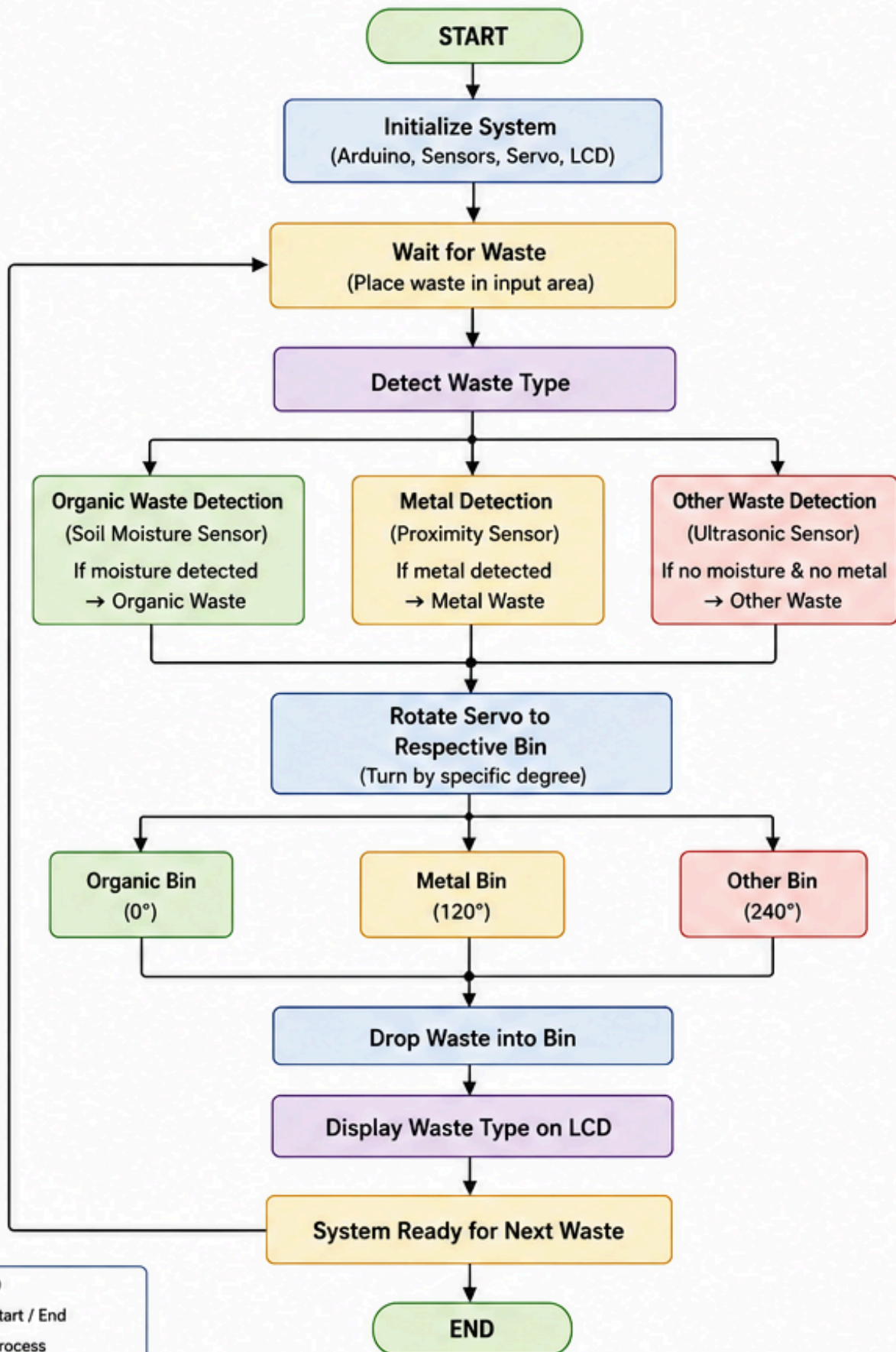
CONNECTION SUMMARY		
Component	Arduino Pin	Description
Soil Moisture Sensor AO	A0	Detects moisture for organic waste
Proximity Sensor DO	A1	Detects metal
Ultrasonic Sensor TRIG	A2	Trigger pulse
Ultrasonic Sensor ECHO	A3	Echo received
LCD RS	D12	LCD Register Select
LCD E	D11	LCD Enable
LCD D4	D5	LCD Data 4
LCD D5	D4	LCD Data 5
LCD D6	D3	LCD Data 6
LCD D7	D2	LCD Data 7
LCD RW	GND	LCD Read/Write (GND)
Servo Signal	D9	Controls servo rotation
All VCC	5V	Power Supply
All GND	GND	Ground



NOTES:

1. Soil Moisture Sensor detects organic waste (moisture).
2. Proximity Sensor detects metal.
3. If neither organic nor metal is detected, remaining waste is treated as other.
4. Servo rotates to 0°, 120° or 240° to align respective bin.
5. LCD displays the detected waste type.

FLOW CHART – JARVIS SMART WASTE SEGREGATION SYSTEM

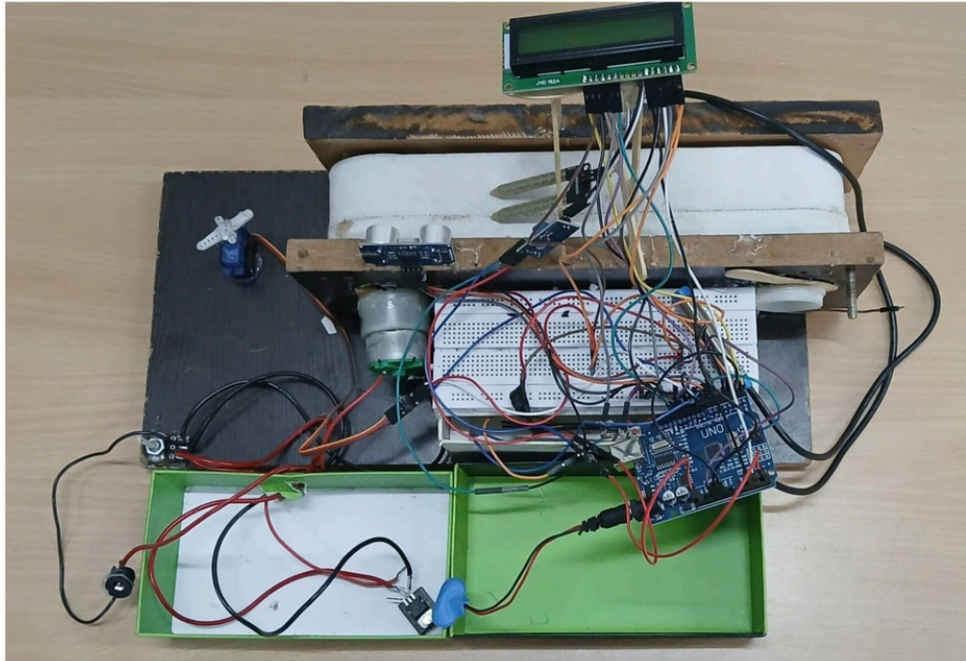


LEGEND

- █ Start / End
- █ Process
- █ Detection
- █ Action / Display
- █ Decision / Output

Prototype Image Analysis

The prototype consists of an Arduino Uno controller, breadboard-based interfacing, LCD display for status output, servo mechanism for directing waste, sensor modules for object detection/classification and collection trays. The architecture demonstrates proof-of-concept functionality for automated segregation.



Component Explanation

LCDDisplay: User feedback and status indication. Arduino Uno: Central controller. Servo Motor: Directs waste into designated bins. Sensors: Detect material characteristics. Breadboard: Rapid prototyping platform. Collection Bins: Store segregated waste categories.

Sample Arduino Code

```
/*
=====
JARVIS
(Just A Rather Very Intelligent Sorting Machine)

Smart Waste Segregation System
Developed By: STARK INNOVATIONS

Hardware:
1. Arduino UNO
2. Soil Moisture Sensor
3. Inductive Proximity Sensor
4. Ultrasonic Sensor HC-SR04
5. Servo Motor SG90
6. 16x2 LCD Display

=====
*/

#include <Servo.h>
#include <LiquidCrystal.h>

// LCD Pins
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

// Sensor Pins
const int moisturePin = A0;
const int proximityPin = 7;

// Ultrasonic Pins
const int trigPin = 8;
const int echoPin = 9;

// Servo Pin
const int servoPin = 10;
```

```
// Servo Object
Servo sortingServo;

// Threshold Values
const int moistureThreshold = 500;
const int distanceThreshold = 15;

// Servo Positions
const int ORGANIC_BIN = 0;
const int METAL_BIN = 90;
const int OTHER_BIN = 180;

// Variables
long duration;
int distance;
int moistureValue;
int metalDetected;

String wasteType = "";

void setup()
{
    Serial.begin(9600);

    pinMode(proximityPin, INPUT);

    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);

    sortingServo.attach(servoPin);

    lcd.begin(16, 2);

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("JARVIS SYSTEM");
    lcd.setCursor(0,1);
    lcd.print("Initializing");

    delay(3000);

    sortingServo.write(90);

    Serial.println("System Started");
}
```

```

void loop()
{
    displayReadyMessage();

    moistureValue = analogRead(moisturePin);

    metalDetected = digitalRead(proximityPin);

    distance = getDistance();

    Serial.print("Moisture: ");
    Serial.println(moistureValue);

    Serial.print("Metal: ");
    Serial.println(metalDetected);

    Serial.print("Distance: ");
    Serial.println(distance);

    if (moistureValue > moistureThreshold)
    {
        processOrganicWaste();
    }

    else if (metalDetected == HIGH)
    {
        processMetalWaste();
    }

    else if (distance < distanceThreshold)
    {
        processOtherWaste();
    }

    delay(1000);
}

//=====
// Ultrasonic Distance Function
//=====
int getDistance()
{

```

```

digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);

    distance = duration * 0.034 / 2;

    return distance;
}

//=====
// Organic Waste Processing
//=====
void processOrganicWaste()
{
    wasteType = "ORGANIC";

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Waste:");
    lcd.print(wasteType);

    lcd.setCursor(0,1);
    lcd.print("Bin -> Organic");

    Serial.println("Organic Waste Detected");
    sortingServo.write(ORGANIC_BIN);

    delay(3000);
    sortingServo.write(90);
    Serial.println("Sorting Complete");
}

//=====
// Metal Waste Processing
//=====
void processMetalWaste()
{
    wasteType = "METAL";

```



```

lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Waste:");
  lcd.print(wasteType);
  lcd.setCursor(0,1);
  lcd.print("Bin -> Metal");
  Serial.println("Metal Waste Detected");
  sortingServo.write(METAL_BIN);
  delay(3000);
  sortingServo.write(90);
  Serial.println("Sorting Complete");
}
//=====
// Other Waste Processing
//=====
void processOtherWaste()
{
  wasteType = "OTHER";
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Waste:");
  lcd.print(wasteType);

  lcd.setCursor(0,1);
  lcd.print("Bin -> Other");

  Serial.println("Other Waste Detected");

  sortingServo.write(OTHER_BIN);

  delay(3000);

  sortingServo.write(90);

  Serial.println("Sorting Complete");
}

//=====
// LCD Ready Message
//=====
void displayReadyMessage()
{
  lcd.clear();

```

```
lcd.setCursor(0,0);  
  lcd.print("JARVIS READY");  
  
  lcd.setCursor(0,1);  
  lcd.print("Place Waste");  
}
```